

**AMENDMENT NO. 1 JANUARY 2009
TO
IS 7224 : 2006 IODIZED SALT, VACUUM EVAPORATED
IODIZED SALT AND REFINED IODIZED SALT —
SPECIFICATION**

(Second Revision)

(Page 9, Annex K) — Substitute the following for the existing:

ANNEX K
[Table 1, Sl No. (x)]

DETERMINATION OF LEAD

K-1 Two methods have been specified for determination of lead - the chemical method and the atomic absorption spectrophotometric method. The atomic absorption spectrophotometric method shall be taken as referee method.

K-2 CHEMICAL METHOD

K-2.1 Apparatus

K-2.1.1 *Nessler Cylinders*, 50-ml capacity

K-2.1.2 Reagents

K-2.1.2.1 *Acetic acid*, approximately 33 percent (w/v).

K-2.1.2.2 *Dilute ammonium hydroxide*, approximately 4 N.

K-2.1.2.3 *Potassium cyanide solution* — Dissolve 10 g of potassium cyanide in 90 ml of water, add 2 ml of hydrogen peroxide (20 volume strength), allow to stand for 24 h and make up to 100 ml with water.

K-2.1.2.4 *Sodium sulphide solution* — Dissolve 10 g of sodium sulphide ($\text{Na}_2\text{S} \cdot 9\text{H}_2\text{O}$) in 100 ml of water.

K-2.1.2.5 *Standard lead solution* — Dissolve 0.160 g of lead nitrate in 5 ml of concentrated nitric acid (conforming to IS 264) and dilute to 100 ml in a graduated flask. Again dilute 10 ml of the solution to 1 000 ml. One millilitre of the solution finally obtained contains 0.01 mg of lead (as Pb).

K-2.1.3 Procedure

K-2.1.3.1 Dissolve 4 g of the dried sample (see C-1) in water in a Nessler cylinder and add 5 ml of acetic acid. Make the mixture alkaline with dilute

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ammonium hydroxide and add 1 ml of potassium cyanide solution. If turbid, filter. Add two drops of sodium sulphide solution and mix well. Carry out a control test in another Nessler cylinder in exactly the same manner but using 1 ml of standard lead solution in place of the dried sample. Dilute the solution in both the cylinders to 50-ml mark. Compare the colour produced in the two cylinders against a white background.

K-2.1.3.2 The material shall be taken to have not exceeded the limit specified in Table 1 if the intensity of colour obtained with the material is not greater than that obtained in the control test.

K-3 ATOMIC ABSORPTION SPECTROPHOTOMETRIC METHOD

K-3.1 Principle

Test portions are dried and then ashed at 450°C under a gradual increase (about 50°C/h) in temperature, 6 N HCl (1+1) is added and the solution is evaporated to dryness. The residue is dissolved in 0.1N HNO₃ and the analyte is determined by flame and graphite procedures.

K-3.2 Apparatus

K-3.2.1 *Atomic Absorption Spectrophotometer*, with an air-acetylene burner or nitrous oxide-acetylene burner for flame and a graphite furnace for electrothermal determinations with appropriate background (non-atomic correction). Instrument parameters are usually given by the manufacturer in the manual provided with the instrument.

K-3.2.2 Hollow cathode or electrodeless discharge lamps for all elements.

K-3.2.3 *Furnace*, programmable or muffle furnace with thermostat maintaining 450 ± 25°C.

K-3.2.4 *Hot Plate*, with heating control to heat up to 300°C.

K-3.2.5 *Quartz or Platinum Dishes*

K-3.2.6 *Polystyrene Bottles*, with leak proof closures – 100 ml

NOTE — Carefully clean and rinse all glassware and plastic ware with HNO₃ or HCl to avoid metal contamination. First wash with water and detergent, rinse with tap water, followed by distilled water, then with dilute acid (1 + 9) and finally 3-4 times with distilled water.

K-3.3 Reagents

K-3.3.1 *Water*, redistilled or deionized

K-3.3.2 *20 Percent Sulphuric Acid*

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K-3.3.3 Hydrochloric Acid A.R. (6N) — Dilute 500 ml hydrochloric acid to one litre with water.

K-3.3.4 Nitric Acid A.R. 0.1M — Dilute 7 ml concentrated acid to one litre.

K-3.3.5 Nitric Acid Concentrated (Sp. Grade 1.40)

K-3.3.6 Lead Standard Solution — 1 mg/ml — Dissolve 1.000 g Pb in 7 ml concentrated HNO_3 in one litre volumetric flask. Dilute to volume with water. Commercially available standard solutions for AAS may be used for all metal standards.

K-3.3.7 Working Standard Solution — For graphite furnace analysis dilute standard solutions with 0.1 M HNO_3 to a range of standards that cover the linear range of the elements to be determined. For Flame analysis dilute standard solutions with 0.1 M HNO_3 to a range of standards that covers the concentration of the elements to be determined.

K-3.4 Procedure

K-3.4.1 Preparation of Sample

Weigh accurately about 25 g of well homogenized sample into a clean silica dish. Add 25 ml of 20 percent sulphuric acid. Mix thoroughly with a glass stirring rod ensuring all sample material is wetted by the acid. Rinse stirring rod with water into silica dish. Dry the contents of the dish thoroughly on a steam bath or in an oven around 110°C. When the sample is thoroughly dried, heat the contents of the dish with a soft flame (such as that of a Bunsen burner) until all volatile or readily combustible matter has been removed. Transfer the dish to a furnace set at 250°C. Slowly raise temperature to 500°C. Ash at this temperature for about 6 to 8 hours. Remove the dish and cool. Ash should now be white or brownish red and essentially be carbon free. If ash contains carbon particles, wash down sides of dish with water and add 2 ml of HNO_3 and mix well. Dry thoroughly on hot plate. Return dish to furnace at 500°C and ash for 30 min. Repeat nitric acid treatment using 1 ml increments of HNO_3 until white/brownish red, carbon free ash is obtained. When clean ash is obtained, remove the dish from furnace, cool and add 1 ml HNO_3 and 10 ml of water. Heat on hot plate till sample ash is dissolved. Quantitatively transfer the contents of the dish to a 100-ml volumetric flask, heat the dish with 10 ml of HCl (1+1) and transfer the solution again to the same volumetric flask to volume with water.

Prepare sample blank solution by following the same procedure as described for sample. Use same quantities of reagents including water for both sample and

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Blank: Subject both sample and sample blank to identical treatment (even the length of time kept in furnace etc.)

NOTES

1 Do not ash HNO_3 in furnace. Always dry HNO_3 (in the dish) on steam bath or hot plate and then ash in furnace.

2 Do not allow sample to ignite during any stage of ashing.

3 If the calcium content of the sample is high, then avoid the use of sulphonic acid (Ash aid) and ash at temperatures not exceeding 450°C .

K-3.4.2 Determination

Atomic Absorption Spectrophotometry — Lead in foods generally requires graphite furnace AAS (GFAAS) for determination.

- a) Set the instrument as per the previously established optimum conditions/as per the guidelines given in the Instruction Manual (provided along with the instrument). The standard conditions for Atomic Absorption Spectrophotometer are wavelength of 217.0 and air – acetylene as flame-gases.
- b) Determine absorbance of sample solution(s) and blank.
- c) Calculate the heavy metal content from standard curve.

K-3.4.3 Preparation of Standard Curve

Read the absorbance of a series of standard metal solutions in the Atomic Absorption Spectrophotometer after setting the instrument as per optimum conditions. Plot absorbance against μg of metal/ml solution.

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